SOUTH CENTRAL RAILWAY

Safety.387/Fly Leaf/14/2025

Fly Leaf No. 14/ 2025

Attention...... All Concerned .....

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As per 6th Correction slip of LHB Maintenance Manual version 1.1 July 2022 Volume- II, (Lt.No: IRCAMTECH/M/GWL/Check-Sheet/ LHB Bogies Date: 07.10.2024, Annexure-B)

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AIR BRAKE TESTING (SINGLE CAR)
(LHB COACHING STOCK)

I. General Instructions to be followed before embarking on Air Brake Testing (Single Car)

a) Air Compressor and Connections.

a. Ensure that the Proper working of Air Compressor.
b. Ensure UT and HT Testing details of Air Reservoir.
c. Ensure Air Drier and Drain Valve connections are in working and operational condition.
d. Ensure that the all connecting line/pipe line joints are free from leakages from Air Compressor Out let to SCTR inlet.

b) Single Car Test Rig (SCTR).

a. Ensure SCTR Connections, Valves, Driver's Brake Valve (DBV) and connections are free from leakages.
b. Ensure Pressure Gauges provided are Calibrated and error free.
c. Working Procedure/Working Instructions to be displayed/ painted at convenient/ readable location.

c) Necessary Tools & Plants.

a. Required nos. of Precision Pressure Gauges (CR, AR & BC) with End Test Points and with connecting hose to be provided.
b. Necessary Hand tools as per IRCAMTECH manual / firm manual to be procured and used.

d) Awareness Programmes.

a. Necessary awareness programmes /classes/meetings to be conducted to update on latest modifications/guide lines/ working procedures.

II. Air Brake Testing (Single Car / SCTR).

(6th Correction slip of LHB Maintenance Manual version 1.1 July 2022 Volume- II)

Air Brake Testing (Single Car)
(Revised Annexure B of Chapter 3)

Coach No		**Type**		**Date of Testing**	
Shop Sch	**I II III**	**Date**		**Return Date**	
DV No		**DV Make**		**Brake Equipment**	
WSP make		**Caliper make**			

Specified Pressure	**BP**	5.0±0.1 kg/cm^2	**Actual**	
	FP	6.0±0.1 kg/cm^2		

Pre-Inspection: Please ensure that all the Pipe fittings, Brake Equipment's are properly fitted and in place before starting of testing.

Item	Test	Specified Value	Actual Value	Remarks
1.0	**Reservoir Charging**			
1.1	Charging Time of AR (0 to 4.8 kg/cm^2)	**175± 30 sec. (SAB/FT)** **110 to 240 sec. (KB)** **165 to 235 sec. (Escorts)** **160 to 200 sec. (JWL DAKO)**		
1.2	Charging Time of CR (6.0 Lt) (0 to 4.8 kg/cm^2)	**165±20 sec. (SAB/FT)** **160 to 210 sec. (KB)** **160 to 210 sec. (Escorts)** **160 to 200 sec. (JWL DAKO)**		
1.3	BP Pressure	**5.0±0.10 kg/cm^2**		
1.4	CR Pressure	**5.0±0.10 kg/cm^2**		
1.5	FP Pressure	**6.0±0.10 kg/cm^2**		
2.0	**Sealing Test** (Allow the system to settle for 2 min. After charging BP & FP. Observe the rate of leakage).			
2.1	BP (Less than 0.1 kg/cm^2 in 5min)	**BP<0.02 kg/cm^2/minute**		
2.2	FP (Less than 0.1 kg/cm^2 in 5min)	**FP<0.02 kg/cm^2/minute**		
3.0	**Full Brake Application**			
3.1	Reduce BP from 5.0 to 3.4 kg/cm^2	**3 – 5 Sec**		
3.2	Brake Accelerator should not respond	**Should not Respond**		
3.3	Maximum BC pressure	**3.0 ±0.1 kg/cm^2**		
3.4	Leakage in BC Pressure within 5 minutes	**<0.1 kg/cm^2**		
3.5	All Brake Cylinders are applied	**Applied**		

3.6	Both side Brake Indicators should show Red	**Red**		
4.0	**Release Full Brake Application**			
4.1	Charge BP (up to 5.0 kg/cm^2)	**5.0±0.1 kg/cm^2**		
4.2	All Brake Cylinders are Released	**Released**		
4.3	Both Side Brake Indicators should show Green	**Green**		
5.0	**Over Charge Protection** Over charge Brake Pipe (BP) to 6 kg/cm^2 after full-service Brake Application.	**Control reservoir pressure should not rise within 10 sec.**		
6.0	**Emergency Application**			
6.1	Reduce BP to 0 kg/cm^2	**0 kg/cm^2**		
6.2	Brake Accelerator should respond	**Blast of Air**		
6.3	Charging time of brake cylinder (0 – 3.6 kg/cm^2)	**3 – 5 Sec.**		
6.4	Max. brake cylinder pressure	**3.0 ±0.1 kg/cm^2**		
6.5	All Brake Cylinders applied	**Applied**		
6.6	Both side Brake indicator window should show red	**Red**		
7.0	**Release emergency Brake application**			
7.1	BC release time (Max to 0.4 kg/cm^2)	**15-20 Seconds**		
7.2	All Brake Cylinder released	**Release**		
7.3	Both side Brake indicator window should show Green	**Green**		
8.0	**Graduated brake application and Release** Graduated brake application and Release (Minimum 7 Steps)	**Brake should apply and release corresponding to decrease and increase of BP Pressure**		
9.0	**Test for Pressure switch for Anti-skid device.**			
9.1	Charge the Feed pipe/Brake pipe* pressure	**OK**		
9.2	Anti-skid device get power supply at 1.8±0.2 kg/cm^2	**OK**		
9.3	Anti-skid device get power supply off at 1.3±0.2 kg/cm^2 *for SAB – FP and for KB – BP	**OK**		
10.0	**Isolation Test**			
10.1	Close the isolating cocks for Bogie – 1 & 2	**Brake should not apply**		
10.2	Reduce BP pressure to full brake application (Brake should not apply)			
10.3	Both side Brake indicators shows Green	**Green**		
10.4	Open both isolating cock (Brake should apply corresponding to opening of isolating cock for bogies)	**Brake Applies**		

10.5	Both side Brake indicators shows Red	**Red**		
10.6	Again, close the Isolating cock of bogie 1&2 one by one.	**Brake will Release**		
10.7	Both side Brake indicators of bogie 1 & 2 show Green one by one.	**Green**		
11.0	**Sensitivity Test**			
11.1	Reduce the BP pressure at the rate of 0.6kg/cm^2 in 6 second. when BP is isolated from main reservoir	**Brake should applied within 6 seconds**		
12.0	**Insensitivity Test**			
12.1	Reduce the BP pressure at the rate of 0.3 kg/cm^2 per minute. when BP is isolated from main reservoir	**Brake should not be applied**		
13.0	**Passenger Emergency Pull Box testing**			
13.1	Pull the emergency pull box handle &check KB should be 09 to 12.5 kgs & 11kgs to 23 kgs for Faively & 09 to 12 kgs for Escorts.	**BP Pressure should remain 2.0± 0.2 kg/cm^2**		
13.2	Brake accelerator does respond	**Should Respond**		
13.3	BP Pressure exhaust from emergency Brake valve	**Yes**		
13.4	Indicator Lamp on outside coach glowing	**Yes**		
13.5	Both side Brake indicators shows Red	**Red**		
13.6	After resetting, exhaust from emergency Brake valve is stopped	**Should Stop**		
13.7	Both side Brake indicators shows Green	**Green**		
14.0	**Hand Brake test (Power Car only)**			
14.1	Apply hand brake by means of wheel	**OK**		
14.2	Both side Hand Brake indicators shows Red	**Red**		
14.3	Brake Cylinders provided with hand brake lever are applied	**Applied**		
14.4	Movement of flex ball cable is proper	**Yes**		
14.5	Release hand brake by means of wheel	**OK**		
14.6	Brake should release	**Releases**		
14.7	Both side Hand Brake indicators shows Green	**Green**		
15.0	**Emergency brake by guard van valve (Power car only)**			

15.1	Drop BP Pressure by means of guard valve (Brake Should apply)	**Brake Apply**		
15.2	Brake Accelerator should respond	**Blast of Air, Brake Accelerator responded**		
15.3	Both side Brake indicators shows Red & Hand Brake indicators shows Green	**OK**		
15.4	Reset guard van valve (Brake should release)	**Brakes Releases**		
16.0	**Manual release test** @ Apply full brake application and pull manual release wire of DV, it should be Released in one brief pull of Manual release valve.	**CR & BC Pressure becomes zero and brake releases**		
17.0	**WSP test**			
17.1	Check all Speed sensor air gap between sensor and Phonic wheel by means of Filler gauge.	**0.9 to 1.4 mm**		
17.2	Charge the FP pressure at specified value.	**1.8 ± 0.2 kg/cm^2**		
17.3	Check the WSP Micro Processor activated	**Activated**		
17.4	Check the WSP Micro Processor showing code 99.	**OK**		
17.5	Check the Dump Valve venting by test mode	**Venting one by one in proper sequence**		
18.0	**Clearance between brake disc & brake Pad**	**1.5 mm for Faiveley & Knorr 2 ± 1 for Escorts & JWL DAKO**		
19.0	**Double Decker coach max. BC pressure shall be**	**3.8 kg/cm^2**		
19.1	Double Decker coach BC filling time shall be recorded	**from 0 to 3.6 kg/cm^2**		
20.0	Brakes applied by reducing BP to 3.6 kg/cm^2 and wait for 30 minutes and then BP raise up to 5.0 kg/cm^2. (Ref. RDSO letter no. LHB / Brake Pad dated 21.01.2022, Sub: Burning of Brake Pad due to brake binding in LHB coaches)	**Brake should Release**		

Note: For releasing of CR pressure in DAKO make Brake system, please refer supplier's manual.

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SAFETY ORGANISATION **SOUTH CENTRAL RAILWAY**